Link Path Network Optimization Report

Model Type: **link_path**

Metric	Value
Status	Optimal
Objective Value	$300.00
Total Revenue	$300.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$300.00
Accepted Demands	1 / 3
Rejected Demands	2 / 3
Acceptance Rate	33.3%
Nodes	6
Edges	8
Total Initial Capacity	150.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	C -> D (100.0%)
Number of Highly Utilized Edges (≥90%)	2
Most Capacity Expansion	None
Total Lost Revenue	$280.00
Number of Expanded Edges	0

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge C -> D: 100.0% used
- Edge B -> E: 100.0% used

Accepted Demands & Flow Paths

Demand 0: A -> F

Bandwidth: 30.0 units **Revenue:** $300.0
Revenue/BW: $10.00/unit

Path	Fraction	Bandwidth
A -> C -> D -> F	50.00%	15.00
A -> B -> E -> F	50.00%	15.00

Rejected Demands

Idx	Source	Target	Bandwidth	Revenue ($)
1	A	F	20.0	200.0
2	A	D	10.0	80.0

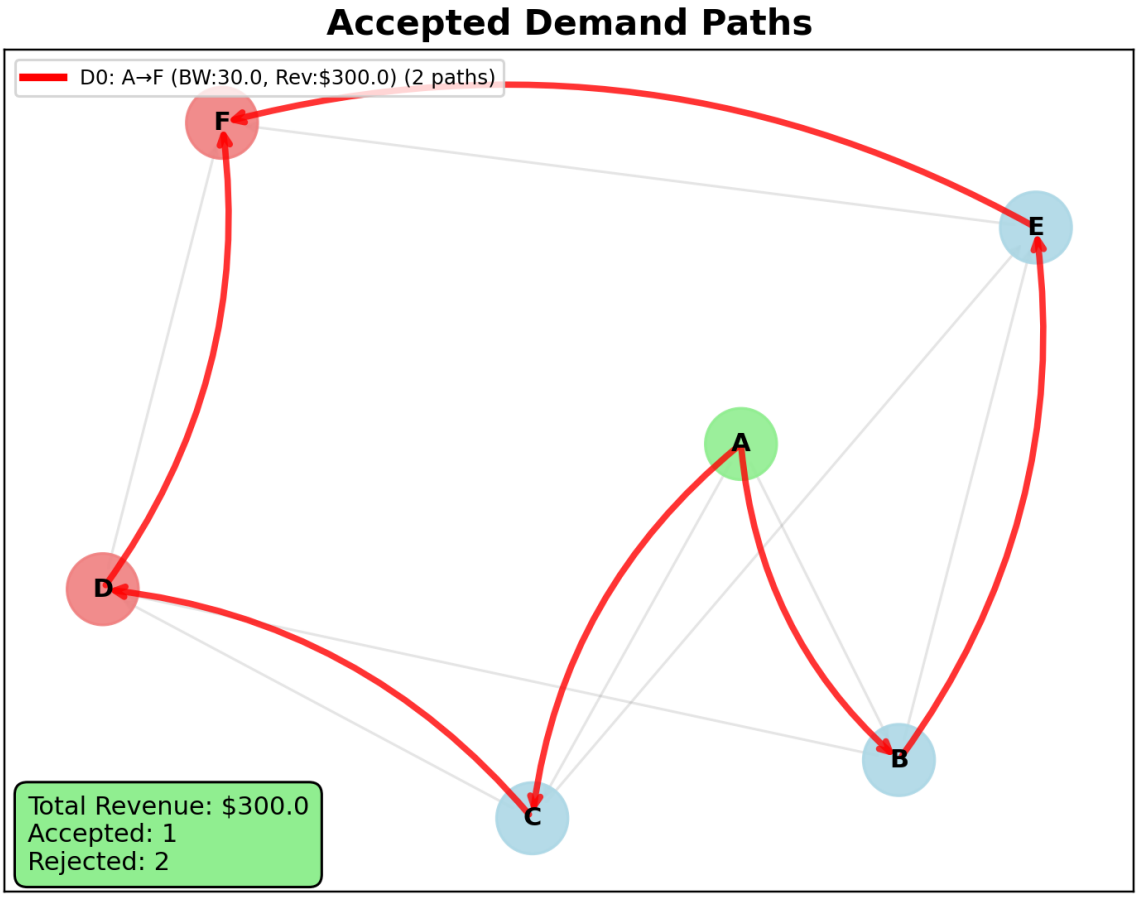
Total Lost Revenue: $280.00

Edge Utilizations (Top 12 Edges)

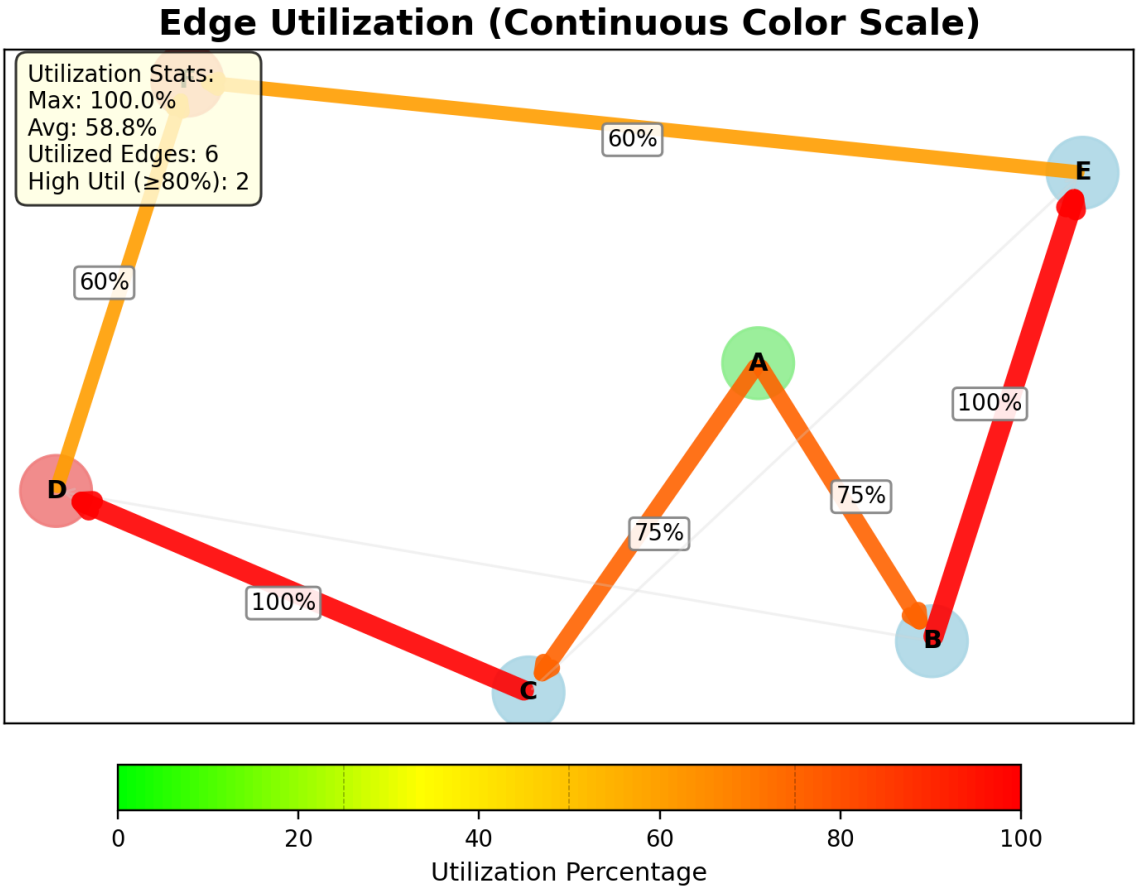
Edge	Flow	Total Capacity	Utilization (%)
C -> D	15.00	15.00	100.0 %
B -> E	15.00	15.00	100.0 %
A -> B	15.00	20.00	75.0 %
A -> C	15.00	20.00	75.0 %
D -> F	15.00	25.00	60.0 %
E -> F	15.00	25.00	60.0 %
B -> D	0.00	15.00	0.0 %
C -> E	0.00	15.00	0.0 %

Network Analysis Visualizations

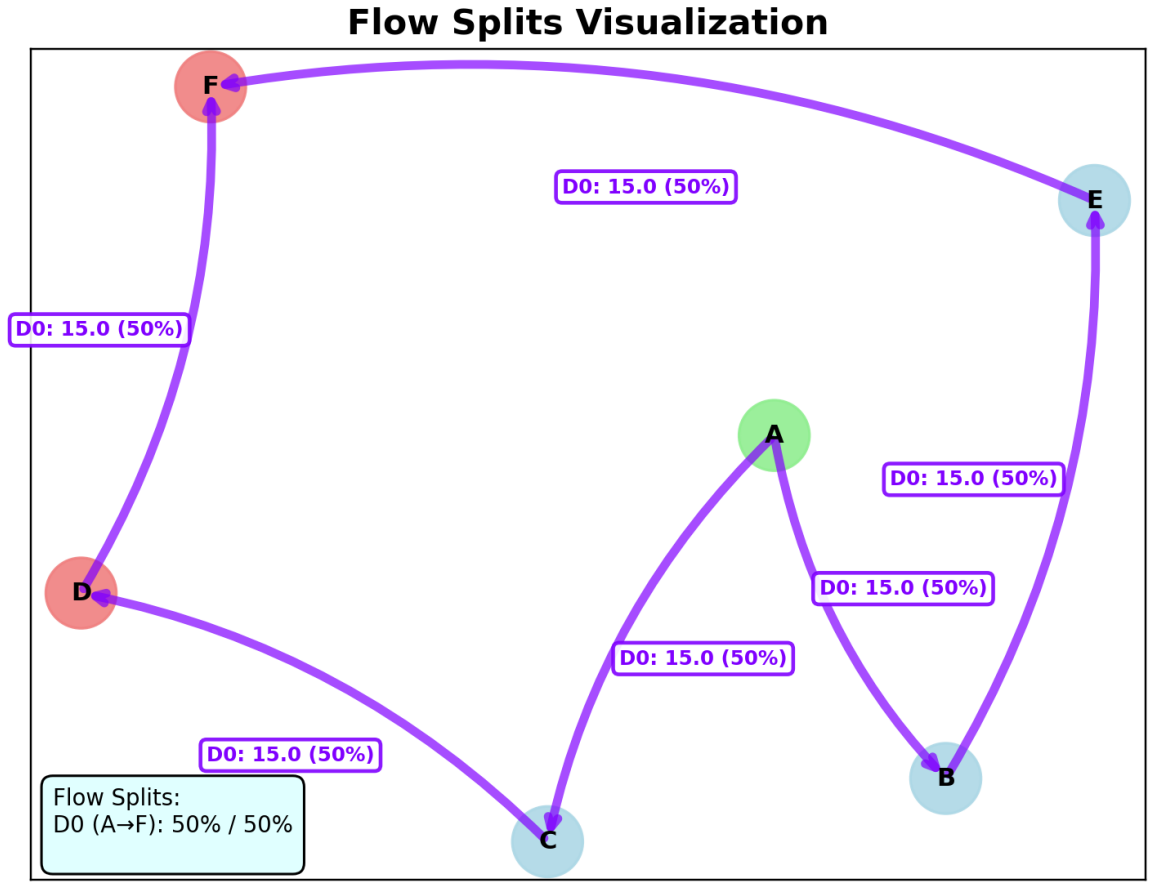
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
F
E
D
A
B
C
D0: 15.0 (50%)
D0: 15.0 (50%)
D0: 15.0 (50%)
D0: 15.0 (50%)
D0: 15.0 (50%)
D0: 15.0 (50%)
Flow Splits:
D0 (A→F): 50% / 50%